

No.	Category	Component Name / Model to Search	Price (EGP)
1	High-Level Computer	Raspberry Pi 4 Model B 8GB	9500
2	Storage	SanDisk Extreme microSDXC 128GB	3200
3	STM32 Development Board	STM32 NUCLEO-F401RE	2100
4	LiDAR Sensor	Slamtec RPLIDAR A1M8	8500
5	Camera	Raspberry Pi Camera Module 3	4500
6	Distance Sensor	VL53L1X Time-of-Flight Sensor	330
7	IMU Sensor	MPU-6050 6-Axis IMU	175
8	Drive Motors	12V DC Geared Motors with Encoders	
9	Motor Drivers	Cytron MDDS30 Motor Driver	
10	Battery	12V LiFePO4 Battery Pack	
11	Battery Management	LiFePO4 Battery Management System (BMS)	
12	Electrical Protection	Automotive Blade Fuse Holder and Fuses	
13	Voltage Regulation	12V-to-5V 5A DC-DC Buck Converter	
14	CAN Transceiver	SN65HVD230 CAN Transceiver	850
15	Raspberry Pi CAN Interface	MCP2515 CAN Controller Module	400
16	Emergency Stop	22mm Mushroom Emergency Stop Switch	240
17	Safety Components	DC-Rated Contactor and Interlock Switch	
18	Robot Chassis	4-Wheel Mobile Robot Chassis / Custom Aluminum Chassis	
19	Wheels	100mm Rubber Robot Wheels	

20	Delivery Compartment	Lockable Metal Storage Box	
21	Access Control	RC522 RFID Reader Module	120
22	Fasteners	M3/M4 Stainless Steel Screw and Nut Kit	
23	Fabrication Materials	Aluminum Sheets and 2020 Aluminum Extrusions	
24	Optional Display	7-Inch HDMI Touchscreen	
25	Cooling and Accessories	5V Cooling Fan and Heatsink	
26	Face Display	7-Inch HDMI LCD Display for Animated Robot Face	4000 /8000
27	Head Enclosure	Custom 3D-Printed Robot Head Shell	
28	Face Camera	Wide-Angle USB Camera	1300
29	Voice Input	USB Microphone Module	
30	Voice Output	Mini Speaker with Audio Amplifier	
31	Humanoid Arms	Custom 3D-Printed Robot Arms	
32	Torso Structure	Custom Robot Torso Enclosure	
33	Neck Structure	Robot Neck Mounting Bracket	
34	Head Movement	Metal Gear Servo Motor with Pan-Tilt Bracket	
36	Display Mounting	Custom LCD Mounting Bracket	
37	Mechanical Assembly	Aluminum Brackets, Hinges, and Mounting Hardware	
39	Power Distribution	Fused Power Distribution Board	